SUMANTH R

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B.Tech. in Computer Science and Engineering (AIML)
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Education

Degree	Institute	CGPA	Duration
B.Tech., Computer Science and Engineering (AIML)	SRM Institute of Science and Technology, Chennai, Tamil Nadu	8.98 / 10	*2023 – 2027*

Experience

- **Works Applications India Pvt. Ltd.** *Dec 2025 – Jan 2026*
  Software Engineering Intern On-site
  - Developed an end-to-end full-stack application using **Java, Spring Boot, MySQL, and React**, implementing **5+ REST APIs** and handling **5K+ simulated requests/day**.
  - Optimized MySQL queries and database indexing, reducing backend response time by **25%** during performance testing.
  - Integrated **Explainable AI** insights into the React interface, improving model interpretability by **20%** and enabling clearer AI-driven outputs.

- **All India Council for Technical Education (AICTE)** *July 2024 – September 2024*
  Google AI/ML Virtual Intern Remote
  - Developed and evaluated **8+ supervised and unsupervised ML models** across **5+ datasets**, achieving up to **92% accuracy**.
  - Applied **data preprocessing, feature engineering, and hyperparameter tuning** across 10+ hands-on labs, improving model performance by **25%**.
  - Implemented **3 bias-mitigation techniques** across AI use cases to improve model fairness and reliability.

Projects

- **Enterprise RAG Knowledge Assistant** *Aug 2026*
  Technologies: Python, FastAPI, LangChain, Hugging Face, FAISS, LLMs
  - Built an end-to-end **RAG pipeline** for document question answering with document ingestion, recursive chunking, embeddings, semantic retrieval, and LLM-based response generation.
  - Implemented **FAISS vector search** with metadata filtering and configurable retrieval parameters to improve context relevance and reduce unsupported responses.
  - Developed a **FastAPI backend** with source citations, conversation history, and evaluation endpoints for measuring retrieval and answer quality.

- **Multi-Agent AI Research Assistant** *Aug 2026*
  Technologies: Python, LLMs, LangGraph, FastAPI, Tool Calling
  - Developed a **multi-agent workflow** with planner, researcher, and synthesis agents to decompose research tasks and generate structured reports.
  - Implemented **tool calling, state management, and agent routing** to dynamically select tools and coordinate multi-step tasks.
  - Added response evaluation and failure handling to identify unsupported claims, incomplete research, and low-confidence outputs.

- **AI Workflow Automation Platform** *Aug 2026*
  Technologies: Python, FastAPI, React.js, LLMs, PostgreSQL
  - Built an AI workflow platform that converts natural-language instructions into executable **LLM-powered workflows** and tool actions.
  - Implemented reusable prompt templates, structured outputs, function calling, workflow state, and execution logging through REST APIs.
  - Developed a **React dashboard** for creating, executing, and monitoring AI workflows with configurable models and prompts.

Publications

- **Deep Learning Based Brain Tumor Classification using Orange Data Analytics** *2026*
  IEEE International Conference (SCEECS 2026) View Paper
  - Developed an end-to-end **deep learning pipeline** using a **16,000+ MRI image dataset** aggregated from Kaggle, BRISC, and FigShare.
  - Achieved **96%+ accuracy** and **0.997 AUC** using Neural Networks with stratified 10-fold cross-validation.
  - Applied **PCA and data augmentation**, reducing feature dimensionality by **75%** while retaining **95% variance** and improving performance by **7% over baselines**.

Achievements

- Selected as a Software Engineering Fellow at Headstarter AI for hands-on AI and software engineering projects.
- Secured 3rd Place in CodeMavens 2025 Hackathon organized by ACM-W SRM Student Chapter.

Technical Skills

- **Generative AI & LLMs**: LLMs, Prompt Engineering, RAG, LangChain, AI Agents, Hugging Face, Vector Databases, Generative AI
- **AI & Machine Learning**: Machine Learning, Deep Learning, NLP, Explainable AI, BERT, RoBERTa, PyTorch, TensorFlow, Scikit-learn
- **AI Automation**: n8n, AI Workflows, Workflow Automation, API Integration, Automated Data Processing, AI-assisted Applications
- **Programming**: Python, Java, C++, C, JavaScript, SQL
- **Development**: React.js, Node.js, REST APIs, FastAPI, Git, GitHub, Linux
- **Data & Databases**: Pandas, NumPy, MySQL, MongoDB
- **Core Concepts**: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks